



React.js Syllabus

1. Introduction to React

- What is React?
- History and evolution of React
- Features of React
- Advantages of using React
- Understanding Single Page Applications (SPA)

2. Prerequisites

- HTML, CSS basics
- JavaScript fundamentals (ES6+)
 - Arrow functions
 - Destructuring
 - Spread & Rest operators
 - Modules (import/export)
- Basic understanding of DOM

3. Setting Up the Environment

- Installing Node.js and npm
- Creating React app using Create React App
- Project structure overview
- Running the development server

4. JSX (JavaScript XML)

- What is JSX?
- Syntax and rules
- Embedding expressions in JSX
- JSX vs HTML

5. Components in React

- Functional components
- Class components
- Component structure and naming
- Reusable components

6. Props (Properties)

- Passing data between components
- Default props
- Props vs state
- Props validation

7. State Management

- What is state?
- useState Hook
- Updating state
- Handling multiple states

8. Event Handling

- Handling user events (click, change, submit)
- Synthetic events
- Binding event handlers

9. Conditional Rendering

- if/else conditions
 - Ternary operators
 - Logical && operator
-

10. Lists and Keys

- Rendering lists using map()
 - Importance of keys
 - Handling dynamic data
-

11. Forms in React

- Controlled components
 - Uncontrolled components
 - Form validation
-

12. React Hooks

- Introduction to Hooks
 - useState
 - useEffect
 - useContext
 - useRef
 - Custom Hooks
-

13. Lifecycle Methods (Class Components)

- Mounting phase
 - Updating phase
 - Unmounting phase
-

14. Routing

- Introduction to routing
 - Installing and using React Router
 - Dynamic routes
 - Navigation
-

15. Context API

- What is Context?
 - Creating and using context
 - Avoiding prop drilling
-

16. Styling in React

- CSS stylesheets
 - Inline styling
 - CSS Modules
 - Styled-components
-

17. API Integration

- Fetch API / Axios
 - Handling API responses
 - Loading and error handling
-

18. State Management Libraries

- Introduction to Redux
 - Actions, reducers, store
 - Middleware (Thunk, Saga)
-

19. Performance Optimization

- Memoization (React.memo)
 - useMemo & useCallback
 - Code splitting
-

20. Testing in React

- Unit testing
 - Tools like Jest and React Testing Library
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21. Deployment

- Building production app
 - Hosting (Netlify, Vercel, GitHub Pages)
-

22. Advanced Topics

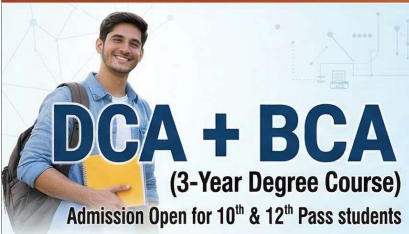
- Higher-Order Components (HOC)
 - Render props
 - Error boundaries
 - Portals
 - Lazy loading & Suspense
-

23. Real-World Projects

- To-do app
 - Blog application
 - E-commerce UI
 - Dashboard
-

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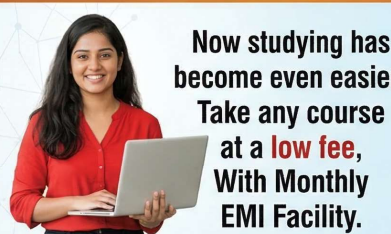
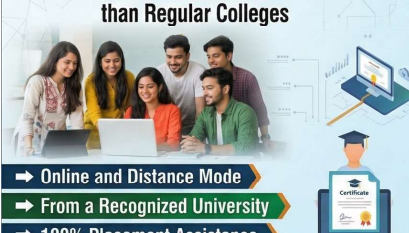
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